

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

```
mrfilosophova@mrfilosophova:~$ touch abc1
mrfilosophova@mrfilosophova:~$
mrfilosophova@mrfilosophova:~$ cp abc1 april
mrfilosophova@mrfilosophova:~$ cp abc1 may
mrfilosophova@mrfilosophova:~$ mkdir monthly
mrfilosophova@mrfilosophova:~$ cp april may monthly
mrfilosophova@mrfilosophova:~$ cp monthly/may monthly/june
mrfilosophova@mrfilosophova:~$ ls monthly
april  june  may
mrfilosophova@mrfilosophova:~$ mkdir monthly.00
mrfilosophova@mrfilosophova:~$ cp -r monthly monthly.00
mrfilosophova@mrfilosophova:~$ cp -r monthly.00 /tmp
mrfilosophova@mrfilosophova:~$
```

Рисунок 1: Выполнение примеров

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ mv april july  
mrfilosophova@mrfilosophova:~$ mv july monthly.00  
mrfilosophova@mrfilosophova:~$ ls monthly.00  
july  monthly  
mrfilosophova@mrfilosophova:~$ mv monthly.00 monthly.01  
mrfilosophova@mrfilosophova:~$ mkdir reports  
mrfilosophova@mrfilosophova:~$ mv monthly.01 reports  
mrfilosophova@mrfilosophova:~$ mv reports/monthly.01 reports/monthly  
mrfilosophova@mrfilosophova:~$
```

Рисунок 2: Выполнение примеров

```
mrfilosophova@mrfilosophova:~$ touch may
mrfilosophova@mrfilosophova:~$ ls -l may
-rw-r--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:29 may
mrfilosophova@mrfilosophova:~$ chmod u+x may
mrfilosophova@mrfilosophova:~$ ls -l may
-rwxr--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:29 may
mrfilosophova@mrfilosophova:~$ chmod u-x may
mrfilosophova@mrfilosophova:~$ ls -l may
-rw-r--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:29 may
mrfilosophova@mrfilosophova:~$ cd
mrfilosophova@mrfilosophova:~$ mkdir monthly
mkdir: cannot create directory 'monthly': File exists
mrfilosophova@mrfilosophova:~$ chmod g-r,o-r monthly
mrfilosophova@mrfilosophova:~$ chmod g+w abc1
mrfilosophova@mrfilosophova:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ cp /usr/include/linux/sysinfo.h ~  
mrfilosophova@mrfilosophova:~$ mv sysinfo.h equipment  
mrfilosophova@mrfilosophova:~$ mkdir ski.plases  
mrfilosophova@mrfilosophova:~$ mv equipment ski.plases/  
mrfilosophova@mrfilosophova:~$ mv ski.plases/equipment ski.plases/equiplist  
mrfilosophova@mrfilosophova:~$ touch abc1  
mrfilosophova@mrfilosophova:~$ cp abc1 ski.plases/equiplist2  
mrfilosophova@mrfilosophova:~$ cd ski.plases/  
mrfilosophova@mrfilosophova:~/ski.plases$ mkdir equipment  
mrfilosophova@mrfilosophova:~/ski.plases$ mv equiplist equipment/  
mrfilosophova@mrfilosophova:~/ski.plases$ mv equiplist2 equipment/  
mrfilosophova@mrfilosophova:~/ski.plases$ cd  
mrfilosophova@mrfilosophova:~$ mkdir newdir  
mrfilosophova@mrfilosophova:~$ mv newdir ski.plases/  
mrfilosophova@mrfilosophova:~$ mv ski.plases/newdir/ ski.plases/plans  
mrfilosophova@mrfilosophova:~$
```

Рисунок 4: Работа с каталогами

Работа с командой chmod

```
mrfilosophova@mrfilosophova:~$ mkdir australia play
mrfilosophova@mrfilosophova:~$ touch my_os feathers
mrfilosophova@mrfilosophova:~$ chmod 744 australia/
mrfilosophova@mrfilosophova:~$ chmod 711 play/
mrfilosophova@mrfilosophova:~$ chmod 544 my_os
mrfilosophova@mrfilosophova:~$ chmod 664 feathers
mrfilosophova@mrfilosophova:~$ ls -l
total 0
-rw-rw-r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:31 abc1
drwxr--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:32 australia
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Desktop
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Documents
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Downloads
-rw-rw-r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:32 feathers
drwxr-xr-x. 1 mrfilosophova mrfilosophova 74 Sep  2 11:02 git-extended
-rw-r--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:29 may
drwx--x--x. 1 mrfilosophova mrfilosophova 24 Sep  2 11:28 monthly
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Music
-r-xr--r--. 1 mrfilosophova mrfilosophova 0 Sep  2 11:32 my_os
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Pictures
drwx--x--x. 1 mrfilosophova mrfilosophova 0 Sep  2 11:32 play
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Public
drwxr-xr-x. 1 mrfilosophova mrfilosophova 14 Sep  2 11:29 reports
drwxr-xr-x. 1 mrfilosophova mrfilosophova 28 Sep  2 11:32 ski.places
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Templates
drwxr-xr-x. 1 mrfilosophova mrfilosophova 0 Sep  2 10:28 Videos
drwxr-xr-x. 1 mrfilosophova mrfilosophova 10 Sep  2 10:38 work
mrfilosophova@mrfilosophova:~$
```

Рисунок 5: Настройка прав доступа

```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:usr/sbin/nologin
polkitd:x:114:114>User for polkitd:/:sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
```

Рисунок 6: Файл /etc/passwd

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ cp feathers file.old  
mrfilosophova@mrfilosophova:~$ mv file.old play/  
mrfilosophova@mrfilosophova:~$ mkdir fun  
mrfilosophova@mrfilosophova:~$ cp -R play/ fun/  
mrfilosophova@mrfilosophova:~$ mv fun/ play/games  
mrfilosophova@mrfilosophova:~$ chmod u-r feathers  
mrfilosophova@mrfilosophova:~$ cat feathers  
cat: feathers: Permission denied  
mrfilosophova@mrfilosophova:~$ cp feathers feathers2  
cp: cannot open 'feathers' for reading: Permission denied  
mrfilosophova@mrfilosophova:~$ chmod u+r feathers  
mrfilosophova@mrfilosophova:~$ chmod u-x play/  
mrfilosophova@mrfilosophova:~$ cd play/  
bash: cd: play/: Permission denied  
mrfilosophova@mrfilosophova:~$ chmod +x play/  
mrfilosophova@mrfilosophova:~$
```

Рисунок 7: Работа с файлами и правами доступа

```
MOUNT(8)                                     System Administration                                     MOUNT(8)
```

NAME

mount - mount a filesystem

SYNOPSIS

mount [-h|-V]

mount [-l] [-t fstype]

mount -a [-fFnrsvw] [-t fstype] [-O optlist]

mount [-fnrsvw] [-o options] device|mountpoint

mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

mount --bind|--rbind|--move olddir newdir

mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION

All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The `mount` command serves to attach the filesystem found on some device to the big file tree. Conversely, the `umount(8)` command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

The standard form of the `mount` command is:

`mount -t type device dir`

This tells the kernel to attach the filesystem found on device (which is of type type) at the directory dir. The option `-t type` is optional. The `mount` command is usually able to detect a filesystem. The root permissions are necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The previous contents (if any) and owner and mode

Manual page mount(8) line 1 (press h for help or q to quit)

```
FSCK(8)                                     System Administration                                     FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a
    device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem
    label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally,
    the fsck program will try to handle filesystems on different physical disk drives in parallel to
    reduce the total amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will
    default to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

    4
        Filesystem errors left uncorrected

    8
        Operational error

Manual page fsck(8) line 1 (press h for help or q to quit)
```

```

MKFS(8)                                     System Administration      MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available under Linux. The filesystem-specific builder is searched for via your PATH environment setting only. Please see the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose
        Produce verbose output, including all filesystem-specific commands that are executed. Specifying this option more than once inhibits execution of any filesystem-specific commands. This is really only useful for testing.

    -h, --help
        Manual page mkfs(8) line 1 (press h for help or q to quit)

```

```
KILL(1)                                User Commands                                KILL(1)
```

NAME

kill - terminate a process

SYNOPSIS

```
kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...
```

```
kill -l [number|0xsigmask] | -L
```

```
kill -d pid
```

DESCRIPTION

The command `kill` sends the specified signal to the specified processes or process groups.

If no signal is specified, the `TERM` signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the `KILL` signal (number 9), since a process may install a handler for the `TERM` signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a `TERM` signal has been sent, then the `KILL` signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

Most modern shells have a builtin `kill` command, with a usage rather similar to that of the command described here. The `--all`, `--pid`, and `--queue` options, and the possibility to specify processes by command name, are local extensions.

If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS

The list of processes to be signaled can be a mixture of names and PIDs.

pid

Each pid can be expressed in one of the following ways:

n

Manual page kill(1) line 1 (press h for help or q to quit)

3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.